Comparative Evaluation of Classification Algorithms on Titanic Dataset

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***Abstract*— On April 15, 1912, the British passenger ship Titanic sank in the North Atlantic, drowning 722 people, including its crew and passengers. Even though this marine accident occurred a long time ago, academics are still interested in finding out which factors contribute to an individual's survival or death. In this study, we propose to apply 7 different machine learning techniques to the publicly available Titanic dataset, including Linear Regression, Logistic Regression (LR), K-Nearest Neighbors (KNN), Ada Boost, Support Vector Machines (SVM), Decision Tree, Random Forest (RF), Gradient Boosting (GB) achieved through machine learning approaches are contrasted with one another. In our study we see that in the methods Logistic Regression give the best accuracy result than other methods. We also use MRMR for feature selection and from this we test and train the data and split the data. By using confusion matrix, we can get a clear result of our predicted result.**

***Keywords*— Titanic, Machine Learning, Classification, Data Analysis, Logistic Regression, Confusion Matrix.**

I. Introduction

The inevitable advancement of technology has improved our quality of life while also posing certain challenges. One advantage of technology is that it makes it simple to get a variety of facts when needed. It is not always feasible to gather the appropriate information, though. It is necessary to process the easily accessible raw data from online sources for it to be useful to an information retrieval system. In this context, feature engineering techniques and machine learning techniques are crucial to the process.

The objective of this study is to use machine learning and feature engineering techniques to obtain the most accurate results from the raw and missing data. Titanic, one of the most widely used datasets in data science, is so used. This dataset catalogs the characteristics of passengers who boarded the Titanic, including who lived and who died. It is acknowledged that the performance of prediction was decreased by some missing and uncorrelated features. The impact of the characteristics has been researched for a thorough data analysis. As a result, certain current features are removed from the dataset and other new characteristics are introduced.

II. Related Works

The Study of Titanic using Machine Learning has garnered significant attention from researchers, resulting in numerous. Research has been conducted in this field. Here are reviews of some of them:

Chatterjee [1] used multiple logistic regression methods and logistic regression. He provided performance figures for several cases. Comparison and found that the highest degree of

accuracy 78.426% of the data from Multiple Linear Regression; the maximum level of accuracy produced by Logistic Regression is 80.756%.

For the Titanic dataset, Datla [2] examined the outcomes of the Decision tree and Random Forests algorithms. Random Forests produced a result of 0.81%, whereas Decision Trees produced a result of 0.84% correctly identified instances. As part of the feature engineering process, they included new variables—such as "survived," "child," "new_fare," "title," "Familysize," and "FamilyIdentity"—that weren't already present in the Titanic dataset. They also substituted the average value of a given feature for any missing values.

Several research that compared several categorization algorithms across various datasets can be found in the literature. Meyer et al.'s [3] comparison of SVM implementation to 16 methods for classification resulted in SVM and neural networks having, respectively, 20% and 21% lower error rates for the titanic dataset. Adaboost classifiers were contrasted with SVM and RBF models by Ratsch et al. [4]. SVM yields a 22.4 percent error rate as the minimal error rate for the Titanic dataset. SVM was utilized by Li et al. [5] as an Adaboost component classifier. The smallest error rate they were able to acquire was 21.8 percent using the Titanic dataset to represent all the experimental sets.

The remainder of the essay is structured as follows: The strategies used in experimental experiments are presented in Section 3. Section 4 provides details on the experimental design and findings. The discussion in Section 5 brings the essay to a close.

III. Methodology

A. Logistic Regression:

A supervised learning classification approach called logistic regression is used to forecast the likelihood of a target variable. One of the simplest ML algorithms, it can be applied to a variety of classification issues, including spam filtering, cancer diagnosis, and other issues. The accuracy score for logistic regression in our investigation is around 92%.

B. Random Forest Classifier:

A popular supervised machine learning approach for classification and regression issues is the Random Forest Classifier. On various samples, it constructs decision trees and uses their average for classification and majority vote for regression. This algorithm's ability to handle data sets with both continuous variables, as in regression, and categorical variables, as in classification, is one of its most significant properties. It produces better outcomes for categorization difficulties. The accuracy score for logistic regression in our investigation is around 94%.

C. Support Vector Machine (SVM):

A machine learning algorithm called Support Vector Machine (SVM) examines data for categorization and regression analysis. It uses supervised learning to examine data and classify it into one of two groups. The sorted data is output as a map with the edges between the two being as far away as possible. It is frequently utilized in the sciences, in handwriting recognition, in picture classification, and in text categorization. The SVM accuracy score in our analysis is almost 96%.

D. K-Nearest Neighbor (KNN):

K-Nearest Neighbor is a classification of data algorithm that looks at the data points nearby to try to establish what category a data point belongs to. Since it does not create a model of the data set in advance, it is essentially a lazy learner algorithm. It only performs calculations when prompted to poll the neighbors of the data points. This makes it very simple to use K-NNN for data mining. The K-Nearest Neighbor score for accuracy in our analysis is almost 96%.

E. Decision Tree:

Among the most popular classifiers is the decision tree because of its relatively easy to construct structure. A decision tree is a model comprising prediction and decision nodes that is tree structured. Prediction nodes provide class labels, whereas decision nodes are utilized to branch. A decision tree is created by C4.5's decision tree algorithm using knowledge gained from training data. C4.5 employs a divide and conquer strategy while creating decision trees.

F. Gradient Boosting:

Gradient Boosting a potent machine learning algorithm created by Friedman (2001), has achieved notable success in a variety of practical applications. In terms of algorithmic optimization in function space, GB handles boosting as a technique for function estimation.

G. Ada Boost:

One of the most popular and successful ensembles learning techniques is AdaBoost. The fundamental idea behind AdaBoost is that by linearly adding several weak classifiers, a strong classifier can be produced. AdaBoost increases the weights of incorrectly categorized data points while decreases the weights of successfully classified data points during the training phase. AdaBoost, then, reweights all data used for training throughout each iteration. Weak classifiers are used in serial application, and the resulting classification models are integrated using weighted majority voting.

IV. Experiments

A. Dataset:

Kaggle contributed to the Titanic Machine Learning from Disaster competition dataset. The Titanic dataset consists of two sets of passengers: a training set with 891 travelers and a test set with 418 passengers who are not included in the training set. Table I provides an explanation of the features.

Table I

Number of features in the Dataset

Feature	Value of Feature	Feature Characteristic
PassengerID	1-1309	Integer
Survived	0,1	Integer
Pclass	1-3	Integer
Name	Name of Passengers	Object
Sex	Male, Female	Object
Age	0-80	Real
SibSp	0-8	Integer
Parch	0-6	Integer
Ticket	Ticket Number	Object
Fare	0-512	Real
Cabin	Cabin Number	Object
Embarked	S, C, Q	Object

Data Analysis:

After reading the dataset “Titanic Dataset” we calculated the summary where the dataset contains 1309 rows and 12 columns. In this data set decision-making column is just integer.

Correlation between data:

Correlation, which is frequently used as a preliminary strategy to uncover correlations between variables in machine learning classification tasks, might be a key to increasing the accuracy of a prediction model. To determine how independent features affect intuitive predicting, classification algorithms might employ the positive or negative correlation between feature values. Figure 4 displays the relationship between the Titanic dataset's characteristics. The embarked and sex traits do not count toward this total because they just have nominal values. When the correlation scores are assessed, it is seen that "Survived" and "Sex" have the highest correlations, whilst "Survived" and "Age" have the lowest correlations. In addition, there is a correlation between the "Sibsp" and "Parch" traits of 0.37. Considering this, a new feature can be produced by combining these two features.

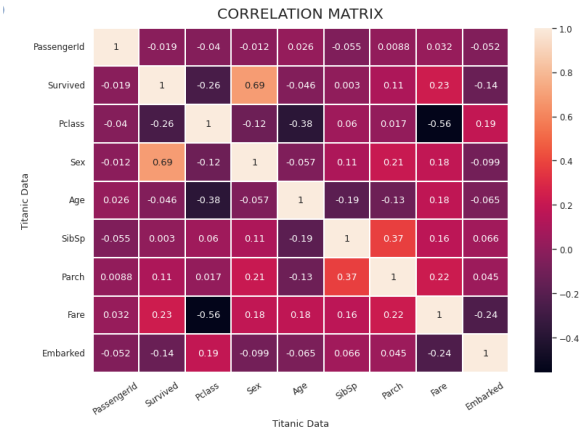

Fig. 4 Correlation between data

Sex:

When we consider the distribution of the “Sex” feature. There are 35.5% of female passengers and 64.5% of male passengers. Here we considered 0 as male and 1 as female. In our study we see that approximately 110 male were survived and approximately 390 female passengers were survived. About approximately 750 male and 90 female were died in this incident. When we examine these distributions, it becomes clear that women have a higher survival rate than men. It has been determined that this attribute has a considerable impact on predicting the class label.

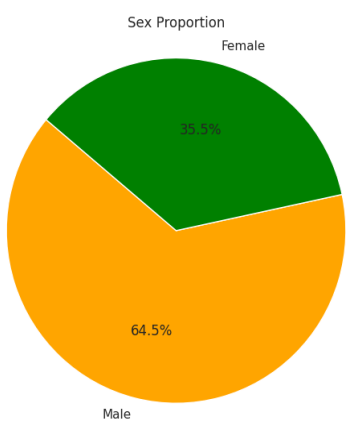

Fig. 1 Sex Proportion

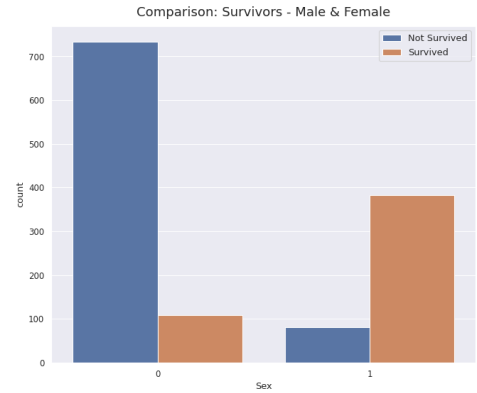

Fig. 2 Distribution of sex features

Pclass:

The "Pclass" feature denotes three distinct passenger classes. There are approximately 330 passengers belongs to class 1, 270 passengers in class 2 and 700 passengers in class 3. Fig. 3 displays the passenger survival rates because of the "Pclass" feature. Class 1 travelers had the best survival statistics. This ratio also demonstrates the existence of affluent people.

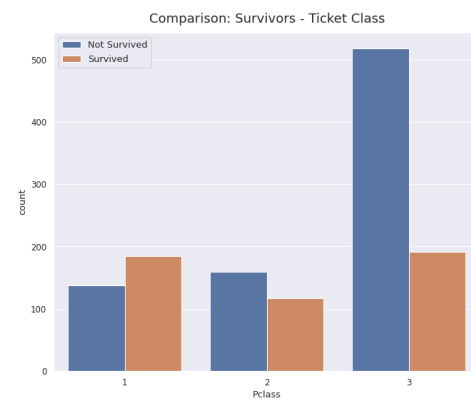

Fig. 3 Distribution of Pclass

Embarked:

According to the distribution of the "Embarked" feature, the ship's ports “Southampton”, “Cherbourd” and “Queenstown” are each home to 270, 125, and 910 people, respectively. Fig. 5 displays the rate of survival rates of travelers embarking from various ports. The port with the most survival rate in this analysis is C. This suggests that the "embarked" feature provides crucial survival information.

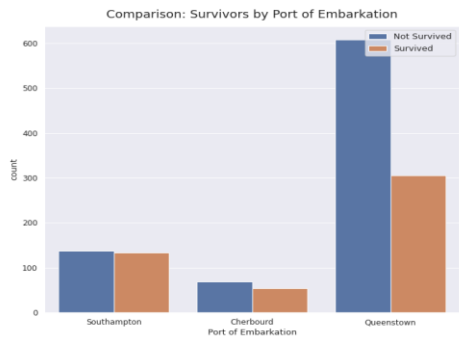

Fig 5 Survival rate of Embarked

MRMR:

By choosing a subset of the most meaningful and non-redundant features from a dataset, feature selection is essential in improving the efficiency of machine learning models. In this study, we use the Titanic dataset, a well-known binary classification problem in the field of data science, to apply the Lowest Redundancy Maximum Relevance (MRMR) feature selection method. The Titanic dataset includes data on the demographics and survival rates of passengers aboard the tragic RMS Titanic.

Building precise and understandable machine learning models requires careful feature selection, especially when working with high-dimensional datasets. The information theory based MRMR technique aims to balance choosing features that are highly relevant to the target variable with reducing redundancy among them. To find a subset of features that can enhance the performance of survival prediction models, we study the use of MRMR on the Titanic dataset in this paper.

By applying MRMR we find five features from our dataset. The result features are age, sex, embarked, sipsb, pclass.

T-SNE:

Techniques for reducing dimensionality are crucial for data processing and visualization, particularly when working with high-dimensional datasets. In this paper, we use the Titanic dataset, a well-known dataset in the field of data science, to apply t-Distributed Stochastic Neighbor Embedding (t-SNE). Our goal is to get insights into the innate patterns and groupings among passengers based on their attributes and lower the dimensionality of the dataset.

A crucial tool in machine learning and statistics used to assess the effectiveness of a classification model is the confusion matrix. It offers a precise and succinct approach to explain how well a model is doing in terms of its capacity to divide data into various groups or classes. The confusion matrix is commonly a square matrix, with the actual and anticipated class labels represented by the rows and columns, respectively.

Dimensionality reduction methods like t-SNE are essential for comprehending complex datasets better. The Titanic dataset includes a variety of passenger characteristics, such as sociodemographic statistics and survival rates. Using t-SNE, we aim to identify latent patterns, clusters, and relationships among passengers by lowering the dimensionality of this dataset.

For make our model unbiased we must scale our data. So, we have used data_norm which is data normalization. basically data_norm feature was the scale data; it has all five parameter but those were scale. Now when we fit transform our data using t-SNE it has converted that 4d data to a single dimension. and we will be plotting x y correlation to see the 2d of our data.

After T-SNE we train and test the features of the dataset and apply the different methods which are suitable for the selected features and done the confusion matrix.

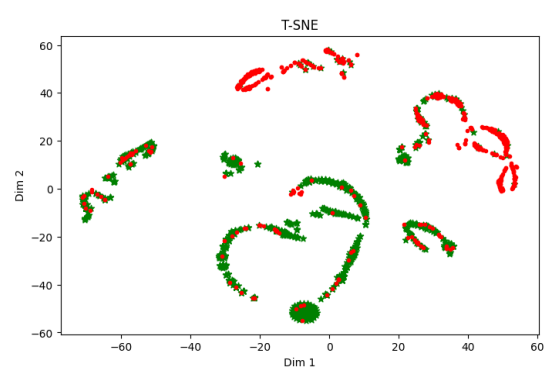

Fig 6 T-SNE

Confusion Matrix:

Random Forest:

In Figure 7 the Random Forest classifier shows the confusion matrix. In the graph the False negative value is 22, True-positive value is 83, False positive value is 135 and True negative value is 22.

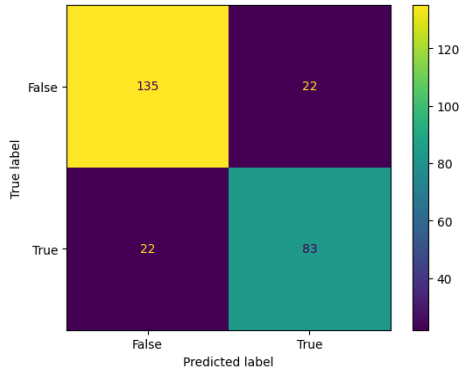

Fig. 7 Random Forest Confusion Matrix

Ada Boost:

In Figure 8 the Ada Boost classifier shows the confusion matrix. In the graph the False negative value is 21, True-positive value is 84, False positive value is 143 and True negative value is 14.

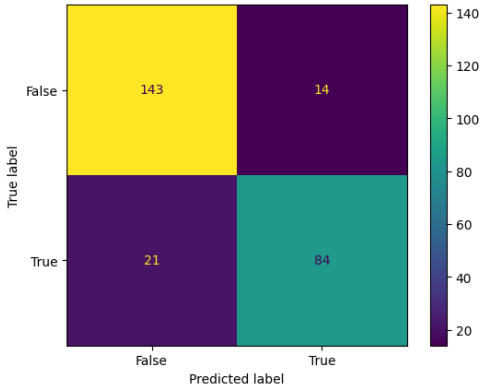

Fig. 8 Ada Boost Confusion Matrix

K-NN Classifier:

In Figure 9 the K-NN classifier shows the confusion matrix. In the graph the False negative value is 33, True-positive value is 72, False positive value is 139 and True negative value is 18.

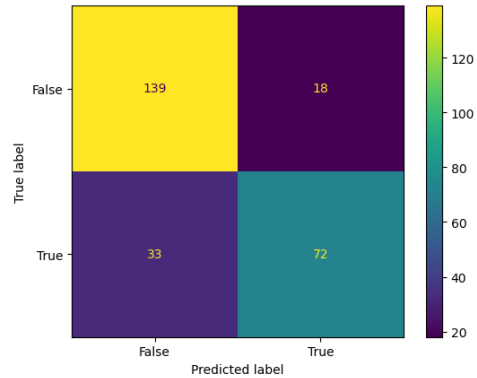

Fig. 9 K-NN Classifier Confusion Matrix

Logistic Regression:

In Figure 10 the Logistic Regression classifier shows the confusion matrix. In the graph the False negative value is 21, True-positive value is 84, False positive value is 144 and True negative value is 13.

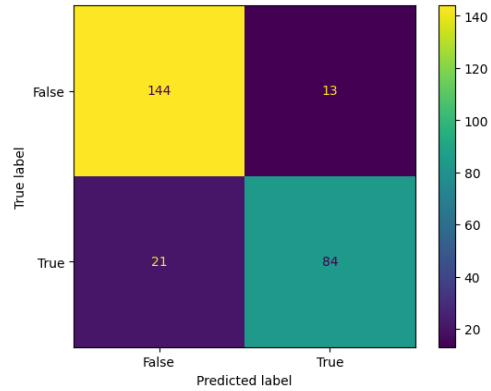

Fig. 10 Logistic Regression Confusion Matrix

Support Vector Machine (SVM):

In Figure 11 the SVM classifier shows the confusion matrix. In the graph the False negative value is 20, True-positive value is 85, False positive value is 143 and True negative value is 14.

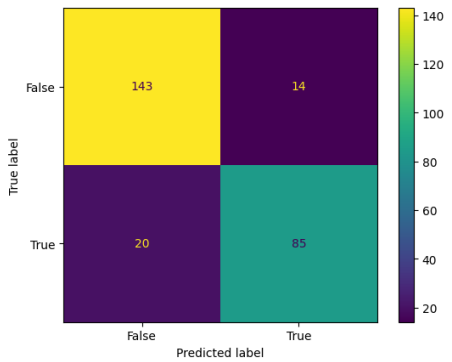

Fig 11 SVM Confusion Matrix

B. Preprocessing Steps:

Data marge, data cleaning, data integration, and data transformation are used as preparation processes in this study. The measurand values of the Age and Fare characteristics are used to fill in the missing data. The "C" value fills in the missing values for the "Embarked" feature. The feature set no longer includes the PassengerId, Name, Ticket, and Cabin features.

C. Experimental Results:

All algorithms are used to examine the chance of survival and discover the characteristics that are related to passenger and crew survival. We discovered that further tweaks to several model parameters are needed to make the algorithm accurate when applying methods to the Titanic dataset.

Table II

Comparison of Accuracy score, precision score, Recall score and F1 score

Algorithm	Accuracy Score	Precision Score	Recall Score	F1 Score
Random Forest	0.83	0.79	0.79	0.79
Ada Boost	0.86	0.86	0.8	0.83
K-NN	0.8	0.8	0.68	0.74
Logistic Regression	0.88	0.86	0.8	0.83
SVM	0.87	0.86	0.81	0.83
Gradient Boosting	0.86	0.86	0.8	0.83

The normal ML strategy and the deep learning to forecast approach were the two approaches used in this study. This research suggests a neural network (ANN) model for the DL technique that is based on artificial neural networks. The improved deep ANN was compared against RF, SVM, LR, and KNN in the conventional ML technique. The crucial features from the database were chosen using the correlation matrix, univariate, and accuracy-precision-recall feature selection approaches. On certain features, both the standard ML models and the optimized deep ANN are used. The findings indicate that for cross-validation and testing outcomes, the optimized deep ANN with chosen features by univariate technique has attained the maximum performance.

VI. Discussion

Through analysis, we were able to pinpoint numerous crucial elements that had a big impact on the Titanic passengers' chances of surviving. Notably, the "Sex" feature's representation of gender emerges as a powerful prediction of survival, with a significantly greater survival rate for females. Passenger class (expressed by "Pclass"), which infers socioeconomic position, demonstrated a clear influence on survival. First-class passengers had a significantly greater survival percentage than passengers in other classes, probably because they had easier access to safety precautions and were closer to lifeboats. There were some variations in the survival rates at the port of embarkation (indicated by "Embarked"). Compared to passengers that boarded in Southampton (S) or Queenstown (Q), Cherbourg (C) passengers had a somewhat greater survival rate. This result could be explained by variations in passenger demographics or cabin settings.

Using the chosen attributes, we created prediction models, and then we assessed their effectiveness using a variety of indicators. Although our models have a respectably high level of accuracy, it is important to understand the limitations of these findings. Class imbalances can have an impact on predictive accuracy, and subsequent research could investigate ways to counteract these biases.

Our study has several drawbacks. First, not all pertinent factors determining survival may not be included in the dataset. Additional data sources or more sophisticated feature engineering techniques might be explored in future study. Future research could look at survival duration or other related outcomes as our study concentrated on binary survival prediction.

We must be aware of the ethical ramifications of such analyses as we explore the variables impacting survival. We must keep in mind that these conclusions are based on a terrible incident even if our goal is to draw insights from historical data. When analyzing and debating the findings, caution and tact should be used.

VII. Conclusion

In a knowledge-based society, it is crucial to get useful results from incomplete and unprocessed data utilizing machine learning and feature engineering techniques. In this essay, we present models for determining if a Titanic tragedy survivor will be identified. To study features that have correlation or are uninformative, a thorough data analysis is first carried out. Additionally, as part of the preprocessing process, marge the train and test dataset, while others are removed, like name, ticket, and cabin. Second, the dataset created in the preparation step is classified using different machine learning algorithms in the classification step. This research compares machine learning methods to evaluate the Titanic dataset to determine which methods are robust and which features influence categorization outcomes.

VIII. Reference

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